Daniela Betancurt-Anzola

Microbial ecology | Multi-omics | Experimental microbiology

danibeta07@gmail.com | dkb5890@psu.edu | Personal website

Microbial ecologist studying how microbial interactions shape community function and responses to disturbance. My doctoral research combines field sampling, genome-resolved metagenomics, and metabolomics to investigate carbon cycling in a neotropical Andean wetland, the páramo. My previous work in gut microbiome research included screening human gut microbial genomes for mercury-resistance genes, engineering bacterial strains, and testing candidate probiotics in mice. I'm interested in using controlled experiments to test hypotheses that emerge from microbiome data.

Education

2022–present **Ph.D. Student, Biochemistry, Microbiology and Molecular Biology**, The Pennsylvania State University, State College, PA

- Expected completion: Summer 2027.

Advisor: Estelle Couradeau, Ph.D. (2024–present).

2021–2022 **M.Sc. Computational Biology**, Universidad de los Andes, Bogotá, Colombia

- Thesis: Divergent response of the gut microbiome to a high-fiber diet.

Advisors: Alejandro Reyes, Ph.D., and Federico Rey, Ph.D.

2012–2018 **B.Sc. Microbiology and B.Sc. Biology; Minor in Psychobiology**, Universidad de los Andes, Bogotá, Colombia

- Thesis: Effects of exposure to *P. aeruginosa* PAO1 on the behavior of zebrafish *Danio rerio*.

Advisors: Veronica Akle, Ph.D., and Martha Vives, Ph.D.

Research Experience

2024–present **Graduate Researcher, Couradeau Lab.**, The Pennsylvania State University, State College, PA

- Designed and lead a multi-omics study of Chingaza páramo soils. Coordinated the collection and processing of 27 soil samples for DNA extraction and untargeted GC-MS/MS and LC-MS/MS metabolomics, as well as permits to transport plant material from Colombia to the United States.
- Developed a metagenomics workflow in Bash, Python, and R using SLURM to process raw reads, reconstruct microbial genomes, estimate strain or genome abundances, and assign taxonomy and gene functions.
- Process untargeted metabolomics data in MZmine and integrate metabolite profiles with shotgun sequencing data, greenhouse-gas fluxes, and soil biogeochemistry. Use multivariate analyses and mixed-effects models to examine relationships across datasets.
- Amplicon data analysis and interpretation using Illumina 16S and ITS sequencing and PacBio LSU sequencing. Examined how plant-microbiome interactions shape microbial communities and how soil microbiomes respond to climate related disturbances.

2022–2024 **Graduate Researcher, Bisanz Lab.**, The Pennsylvania State University, State College, PA

- Examined the prevalence of mercury resistance genes in the human gut by screening the Unified Human Gastrointestinal Protein (UHGP) catalog using a HMMER profile from BacMet reference sequences.
- Engineered *Lacticaseibacillus rhamnosus* strains using homologous recombination and plasmid cloning, including a construct capable of complete methyl-mercury reduction. Quantified its mercury removal *in vitro*.
- Programmed and validated Python protocols for Opentrons liquid-handling robots to screen bacterial strains for methyl-mercury resistance and sequestration.
- Conducted toxicity experiments and performed mouse studies comparing wild-type and engineered *L. rhamnosus*, including study design, dosing, animal handling, sample collection, tissue processing, and analysis.

- Performed strain-specific qPCR on fecal DNA from gnotobiotic mice to quantify bacterial strains, providing an independent benchmark for StrainR2's metagenomic abundance estimates.
- Mentored undergraduate researchers in cloning, managed project timelines, reviewed code and analyses, and provided feedback.

2020–2022 **Research Assistant, Computational Biology and Microbial Ecology Lab.**, Universidad de los Andes, Bogotá, Colombia

- Integrated human gut microbiome amplicon and metabolomics data from a mouse dietary-fiber experiment using QIIME2, R, and Python.

2021–2022 **Laboratory Manager, Affect and Cognition Lab.**, Cornell University, Ithaca, NY

- Developed molecular biology workflows, PCR protocols, genotyped human saliva samples, and analyzed gut metagenomic data in relation to physiological and behavioral measurements.
- Wrote SOPs, protocols, documentation, and training materials while working under IACUC, IRB, biosafety, and controlled-access human-data requirements.
- Mentored undergraduate students, coordinated project timelines and assignments, reviewed analyses, and provided feedback.

2018–2019 **Research Assistant, Affect and Cognition Lab.**, Cornell University, Ithaca, NY

- Completed bench-to-bioinformatics profiling of the human gut microbiome, including DNA extraction, purification, normalization, PCR optimization, and 16S rRNA gene sequencing of human stool samples.
- Trained Long–Evans rats, recorded brain data, and assisted with brain surgeries for studies of memory, aging, and dopamine decline.

Additional Research Experience

2018 **Summer Research Intern, Neurobiology of Memory and Learning Lab.**, Cornell University, Ithaca, NY.

2017–2018 **Undergraduate Researcher, Neuroscience and Circadian Rhythms Lab.**, Universidad de los Andes, Bogotá, Colombia. Evaluated the effect of *P. aeruginosa* PA01 on the behavior of zebrafish using EthoVision.

2013 **Undergraduate Researcher, Mycology and Phytopathology Lab.**, Universidad de Los Andes, Bogotá, Colombia. Developed a protocol for the preparation of permanent fungi demonstrative slides for the laboratory class. Maintained the fungi collection of the lab.

Publications

Betancurt-Anzola, D., Freire-Zapata, V., Villa, J. A., Robles Zaragoza, Cala, M., L., Tfaily, M., Santamaría Vanegas, & Couradeau, E. Deciphering Microbiome–Metabolome Relationships in the Chingaza páramo Through Multi-Omics Integration **(In preparation).**

Betancurt-Anzola, D., Glass, S., J. A., Santamaría Vanegas, & Couradeau, E. Host Identity and Soil–Plant Compartments Structure Microbial Communities in Endemic Páramo Plants **(In preparation).**

Betancurt-Anzola, D., Robles Zaragoza, L., Glass, S., Machado, J.-L., J. A., Santamaría Vanegas, & Couradeau, E. Restructuring of Páramo Soil Microbiomes Under Contrasting Temperature and Moisture Regimes **(In preparation).**

Betancurt-Anzola, D.*, Robles Zaragoza, L.*, Tami-Barrera, L.*, Buytaert, W., Calispa, M., Castillo-Figueroa, D., Diazgranados, M., Espinosa-Ramirez, A. J., Gallery, R. E., Gerard, F., Gutiérrez-Lagoueyte, M. E., Hofstede, R., Hurtado-M, A. B., Lasso, E., Longo Sánchez, M. C., López-Alvarez, D., López-Uribe, M. M., Morales-Rozo, C., Rocha Cárdenas, D. S., Ruiz-Carrascal, D., Sanchez, A., Scott, C. A., Suarez Ballesteros, C. I., Suarez, E., Tovar, C., Villa, J. A., Vuille, M., Rosero-Cuastumal, E. P., Benavides Bolanos, J. A., Santamaría Vanegas, J., Machado, J.-L., Maximova, S., & Couradeau, E. The páramos' invisible collapse: Integrating soil carbon, water, and governance for Andean sustainability. **(In preparation).**

Betancurt-Anzola, D., Vanegas, J., Jurgensen, S. K., Wrighton, K. C., Santamaría-Vanegas, J., & Couradeau, E. (2026). Genome-resolved metagenomics reveals non-methanogenic lineages in an Andean páramo wetland. ISME Communications. doi:10.1093/ismeco/ycag294

Couradeau, E., Vanegas, J., **Betancurt-Anzola, D.**, Glass, S., Eckert, K., Bechtold, E. K., Ellenbogen, J. B., Robles Zaragoza, L., Wrighton, K., & Santamaría Vanegas, J. (2026). Soil microbial diversity in páramos wetland of the

Colombian Andes reveals novel and unique features within a global wetland database. Microbial Ecology. doi:10.1007/s00248-026-02738-x

Heber, K., Tian, S., **Betancurt-Anzola, D.**, Koo, H., & Bisanz, J. E. (2025). StrainR2 accurately deconvolutes strain-level abundances in synthetic microbial communities. Bioinformatics, 41(8), btaf440. doi:10.1093/bioinformatics/btaf440

Pérez-Quintero, A. L., Rodriguez-R, L. M., Cuesta-Morrondo, S., Hakalová, E., **Betancurt-Anzola, D.**, Valera, L. C. C., Cardenas, L. A. C., Matiz-Céron, L., Jacobs, J. M., Roman-Reyna, V., Muñoz, A. R., Giraldo, A. J. B., & Koebnik, R. (2023). Comparative genomics identifies conserved and variable TAL effectors in African strains of the cotton pathogen Xanthomonas citri pv. malvacearum. Phytopathology, 113(8), 1387–1393. doi:10.1094/PHYTO-12-22-0477-SC

*Equal contribution.

Research Presentations

2026 **Poster, One Health Microbiome Center Symposium**, The Pennsylvania State University, University Park, PA.
Abundant uncultivated lineages lacking methanogenic potential dominate carbon-cycling genes in Andean páramo soils

2026 **Talk, Penn State Water Conference,** The Pennsylvania State University, University Park, PA.
Abundant uncultivated lineages lacking methanogenic potential dominate carbon cycling genes in Andean páramo soils

2026 **Poster, JGI User Meeting Integrating Experimentation with Data**, Seattle, WA.
Genomic Dark Matter Drives Metabolic Innovation in High-Altitude Tropical Andean Wetlands

2025 **Poster, Páramos 2025 Symposium**, Bogotá, Colombia.
Microbial Community Structure and Function Along Topographically Driven Microenvironments in a Tropical Andean Peatland; **Best Poster Award**

2025 **Poster, Decoding the Biosphere Symposium**, New England Biolabs, Ipswich, MA.
Microbial Community Structure and Function Along Topographically Driven Microenvironments in a Tropical Andean Peatland

2023 **Rapid-fire talk and poster, ASM Microbe**.
Understanding and Manipulating Dietary Metal Bioavailability through the Gut Microbiome

2020 **Invited Speaker, North-South-South Network Xanthomonads Genomics Course**.
Sequencing Methods An Overview

2018 **Poster, V Biological Sciences Congress**, Universidad de los Andes.
Undergraduate thesis: Effects of the exposure to *P. aeruginosa* PA01 on the behavior of zebrafish *Danio rerio*

Workshop Instruction

2024–2026 **Workshop Chair, Data Analysis Working Group**, One Health Microbiome Center, The Pennsylvania State University. Offered four workshops per semester to the microbiome center community.

2024 **Workshop Lead, National Microbiome Data Collaborative**, One Health Microbiome Center, State College, PA

2024 **Workshop Lead, National Microbiome Data Collaborative**, Joint Genome Institute User Meeting, Walnut Creek, CA

2023 **Teaching Assistant, KickStart Microbiome Workshop**, One Health Microbiome Center, State College, PA

Awards and Fellowships

2025 **Interdisciplinary Innovation Fellowship**, One Health Microbiome Center, The Pennsylvania State University. Awarded $5,000 USD

2025 **Best Poster Award**, Páramos 2025 Symposium, Bogotá, Colombia

2024 **International Graduate Education Award**, The Pennsylvania State University. Supported up to six months of thesis research in the Colombian páramos. Awarded $14,000 USD

2023–2025 **Future Leaders Mentoring Fellowship**, American Society for Microbiology

2018 **Summer Undergraduate Research Fellowship (SURF)**, Universidad de Los Andes. Research placement at Cornell University

Teaching Experience

2023 **Graduate Teaching Assistant, Molecular Genetics Laboratory**, The Pennsylvania State University. Spring and Fall 2023.

2015 **Undergraduate Teaching Assistant, Cell Biology Laboratory**, Universidad de los Andes. Fall 2015.

Leadership and Service

2025–2026 **Organizing Committee Member, One Health Microbiome Center Symposium**, The Pennsylvania State University

2025 **Champion**, National Microbiome Data Collaborative (NMDC)

2024 **Ambassador**, National Microbiome Data Collaborative (NMDC)

2023 **First-Year Representative, Graduate Student Association**, BMMB Program, The Pennsylvania State University

2022 **Microbiome Outreach Instructor**, State College Area High School ESOL Program

Press Releases

2023 **Medical news today.** Could an experimental probiotic help detoxify mercury from diet? https://www.medicalnewstoday.com/articles/could-an-experimental-probiotic-help-detoxify-mercury-from-diet

2023 **American Society for Microbiology** A probiotic could help mitigate mercury absorption in the gut. https://asm.org/Press-Releases/2023/June/How-Microbes-May-Help-Mitigate-Mercury-Absorption

Skills and Methods

- **Programming and reproducible workflows:** R; R Markdown; Python; Bash; Git/GitHub; HPC; SLURM job arrays; Conda; LaTeX.
- **Microbiome bioinformatics:** Genome-resolved metagenomics, including assembly, binning, abundance estimation, taxonomy, and functional annotation. Amplicon data analysis (Illumina 16S and ITS; PacBio LSU). Selected tools: metaSPAdes, CoverM, GTDB-Tk, DRAM2, HMMER, QIIME 2, and DADA2.
- **Metabolomics and statistics:** Untargeted GC-MS/MS and LC-MS/MS metabolomics, MZmine, multi-omics integration, regression, mixed-effects models, and multivariate analysis.
- **Experimental microbiology:** DNA extraction, purification, and normalization; PCR; qPCR; cloning; microbial culture; strain engineering; BONCAT-FACs; flow cytometry; Opentrons Python protocols; liquid-handling automation; high-throughput screening; soil and wetland sampling; conventional and gnotobiotic mouse, rat and zebrafish models.
- **Documentation and compliance:** SOPs; IACUC; IRB; biosafety; controlled-access human data
- **Languages:** Spanish (native); English (advanced); French (intermediate); German (basic)